

Theorem (Bernoulli's inequality). *Let $x \in \mathbb{R}$ satisfy $x \geq -1$ and let $n \in \mathbb{Z}_{\geq 0}$. Then*

$$(1+x)^n \geq 1+nx.$$

Proof. **1. Degenerate and trivial cases.**

Case $n = 0$. $(1+x)^0 = 1 = 1+0 \cdot x$.

Case $x = 0$. $(1+0)^n = 1 = 1+n \cdot 0$.

Case $x = -1$. $(1+x)^n = 0$, while $1+nx = 1-n$. If $n = 1$ we have $0 = 0$; if $n \geq 2$, then $1-n \leq -1$, so $0 \geq 1-n$. Thus the inequality holds for the endpoint $x = -1$.

Henceforth assume $x > -1$ (so $1+x > 0$) and $n \geq 1$.

2. Basis of induction.

For $n = 1$,

$$(1+x)^1 = 1+x = 1+1 \cdot x,$$

so the statement is true (with equality).

3. Inductive hypothesis.

Assume that for a fixed $k \geq 1$ we have

$$(1+x)^k \geq 1+kx \quad \text{for all } x \geq -1. \quad (\text{IH})$$

4. Inductive step.

Multiplying (??) by the non-negative factor $1+x$ yields

$$(1+x)^{k+1} = (1+x)^k(1+x) \geq (1+kx)(1+x). \quad (1)$$

Expanding the right-hand side,

$$(1+kx)(1+x) = 1 + (k+1)x + kx^2. \quad (2)$$

Substituting (??) into (??) gives

$$(1+x)^{k+1} \geq 1 + (k+1)x + kx^2. \quad (3)$$

Since $x^2 \geq 0$ and $k \geq 1$, the term kx^2 is non-negative; discarding it preserves the inequality:

$$(1+x)^{k+1} \geq 1 + (k+1)x.$$

Thus the statement holds for $n = k+1$.

5. Completion of the induction.

The inequality is true for $n = 1$ (basis) and the inductive step shows that truth for $n = k$ implies truth for $n = k+1$. By the principle of mathematical induction it holds for all integers $n \geq 1$. Together with the trivial case $n = 0$ proved in step1, the theorem is valid for every $n \geq 0$.

6. Equality cases.

From (??) we have

$$(1+x)^{k+1} = 1 + (k+1)x \iff kx^2 = 0.$$

Because $k \geq 1$, this forces $x = 0$. Consequently:

- for $n = 0$ equality holds for every admissible x ;
- for $n = 1$ equality holds for every admissible x ;
- for $n \geq 2$ equality holds *iff* $x = 0$.

In compact form,

$$(1+x)^n = 1+nx \iff \begin{cases} x=0, \text{ any } n \geq 0, \\ \text{or } n \in \{0,1\}, \text{ any } x \geq -1. \end{cases}$$

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